PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 1.0 (Introduction)

Report prepared 9 August 2026

1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 1.0 (Introduction) of the review, comprising all 17 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to this review's specific research question, namely concept drift, spatial domain shift, and continual or federated adaptation in machine learning-based IoT intrusion detection, rather than against the clarity of the problem statement in the article's own, different scope. An article can present an impeccably clear problem statement for a different question, such as adversarial robustness, advanced persistent threat detection, or dataset cataloguing, and still score 0 on QC1 here, because that problem statement is not the one this review is screening for.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 1.0 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)

Code	Criterion	Scoring
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 17 full-text articles assessed for Section 1.0 against the quality criteria in Section 2, 10 articles met or exceeded their applicable threshold and were retained, while 7 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 1.0 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security (2020)	2020	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
2	Machine Learning in IoT Security: Current Solutions and Future Challenges (2020)	2020	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
3	Concept Drift Under Harsh Constraints: A Review of Potential Strategies for IoT Systems (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
4	A Taxonomy of Machine-Learning-Based Intrusion Detection Systems for the Internet of Things: A Survey (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
5	A Survey of Deep Learning Technologies for Intrusion Detection in Internet of Things (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
6	A comprehensive survey on deep learning-based intrusion detection systems in Internet of Things (IoT) (2025)	2025	1.0	0.5	1.0	0.5	**3.0**	≥2.5	**Include**
7	Recent endeavors in machine learning-powered intrusion detection systems for the Internet of Things (2024)	2024	1.0	1.0	1.0	0.5	**3.5**	≥2.5	**Include**
8	A Comprehensive Survey on Concept-Drift-Resilient Network Intrusion Detection Systems (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
9	Network Intrusion Detection for IoT Security Based on Learning Techniques (2019)	2019	1.0	1.0	1.0	1.0	**4.0**	≥1.5	**Include**
10	Internet of Things: A survey on machine learning-based intrusion detection approaches (2019)	2019	0.5	0.5	1.0	0.5	**2.5**	≥1.5	**Include**
11	Machine learning based solutions for security of Internet of Things (IoT): A survey (2020)	2020	0.0	0.5	1.0	0.0	**1.5**	≥2.5	**Exclude**
12	Adversarial Machine Learning for Network Intrusion Detection Systems: A Comprehensive Survey (2023)	2023	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
13	A critical review of intrusion detection systems in the internet of things: techniques, deployment strategy, validation strategy, attacks, public datasets and challenges (2021)	2021	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
14	A Review on Machine Learning and Deep Learning Perspectives of IDS for IoT: Recent Updates, Security Issues, and Challenges (2020)	2020	0.0	0.5	1.0	0.0	**1.5**	≥2.5	**Exclude**
15	Network intrusion datasets: A survey, limitations, and recommendations (2025)	2025	0.5	0.5	1.0	0.0	**2.5**	≥2.5	**Exclude**
16	Deep learning for intrusion detection in emerging technologies: a comprehensive survey and new perspectives (2025)	2025	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
17	Machine Learning-Enabled IoT Security: Open Issues and Challenges Under Advanced Persistent Threats (2022)	2022	0.0	0.5	1.0	0.0	**1.5**	≥2.5	**Exclude**

This retained set of 10 articles constitutes the eligible pool for Section 1.0 (Introduction) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

Prepared by
Name: RAZIMAN ZAKARIA